

Foundations of Attention is All You Need

[Attention is All You Need]

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

1.0 Content Block

The idea of encoder–decoder sequence transduction had been developed in the early 2010s; commonly cited as the originators that produced seq2seq are two concurrently published papers from 2014. A 380M-parameter model for machine translation uses two long short-term memories (LSTM). Its architecture consists of two parts. The encoder is an LSTM that takes in a sequence of tokens and turns it into a vector. The decoder is another LSTM that converts the vector into a sequence of tokens. Similarly, another 130M-parameter model used gated recurrent units (GRU) instead of LSTM. Later research showed that GRUs are neither better nor worse than LSTMs for seq2seq. These early seq2seq models had no attention mechanism, and the state vector is accessible only after the last word of the source text was processed. Although in theory such a vector retains the information about the whole original sentence, in practice the information is poorly preserved. This is because the input is processed sequentially by one recurrent network into a fixed-size output vector, which is then processed by another recurrent network into an output. If the input is long, then the output vector would not be able to contain all relevant information, degrading the output. As evidence, reversing the input sentence improved seq2seq translation. The RNN search model introduced an attention mechanism to seq2seq for machine translation to solve the bottleneck problem (of the fixed-size output vector), allowing the model to process long-distance dependencies more easily. The name is because it "emulates searching through a source sentence during decoding a translation". The relative performances were compared between global (that of RNN search) and local (sliding window) attention model architectures for machine translation, finding that mixed attention had higher quality than global attention, while local attention reduced translation time. In 2016, Google Translate was revamped to Google Neural Machine Translation, which replaced the previous model based on statistical machine translation. The new model was a seq2seq model where the encoder and the decoder were both 8 layers of bidirectional LSTM. It took nine months to develop, and it outperformed the statistical approach, which took ten years to develop.

2.0 Content Block

Methods discussed and introduced The paper is best known for introducing the Transformer architecture, which underlies most modern large language models (LLMs). A key reason why the architecture is preferred by most modern LLMs is the parallelizability of the architecture over its predecessors. This ensures that the operations necessary for training can be accelerated on a GPU, allowing both faster training times and models of bigger sizes to be trained. The paper introduced the following mechanisms as part of the development of the transformer architecture.

3.0 Content Block

Dataset - The English-to-German translation model was trained on the 2014 WMT (Workshop on Statistical Machine Translation) English-German dataset, consisting of nearly 4.5 million sentences derived from TED Talks and high-quality news articles. A separate translation model was trained on the much larger 2014 WMT English-French dataset, consisting of 36 million sentences. Both datasets were encoded with byte-pair encoding. Hardware - The models were trained using 8 NVIDIA P100 GPUs. The base models were trained for 100,000 steps, and the big models were trained for 300,000 steps - each

step taking about 0.4 seconds to complete for the base models and 1.0 seconds for the big models. The base model was trained for a total of 12 hours, and the big model was trained for a total of 3.5 days. Both the base and big models outperform the 2017 state-of-the-art in both English-German and English-French, while achieving the comparatively lowest training cost. Hyperparameters and regularization - For their 100M-parameter Transformer model, the authors increased the learning rate linearly for the first 4000 (warmup) steps and decreased it proportionally to the inverse square root of the current step number. Dropout layers were applied to the output of each sub-layer before normalization, the sums of the embeddings, and the positional encodings. The dropout rate was set to 0.1. Label smoothing was applied with a value of 0.1, which "improves accuracy and BLEU score".

4.0 Content Block

$$PE(pos, 2i + 1) = \cos \left(pos / 10000^{2i / d_{model}} \right) \quad \{\displaystyle PE_{\{\rm {pos}\}, 2i+1} = \cos(\rm {pos} / {10000}^{{2i} / d_{\rm {model}}})\}$$

5.0 Content Block

While the primary focus of the paper at the time was to improve machine translation, the paper also discussed the use of the architecture on English Constituency Parsing, both with limited and large-sized datasets, achieving a high-score without specific tuning for the task indicating the promising nature of the model for use in a wide-variety of general purpose of seq2seq tasks.

6.0 Content Block

Predecessors For many years, sequence modelling and generation was done by using plain recurrent neural networks (RNNs). A well-cited early example was the Elman network (1990). In theory, the information from one token can propagate arbitrarily far down the sequence, but in practice the vanishing-gradient problem leaves the model's state at the end of a long sentence without precise, extractable information about preceding tokens. A key breakthrough was LSTM (originally described in a 1995 technical report and formally published in 1997), an RNN that introduced gating mechanisms to mitigate the vanishing gradient problem, allowing efficient learning of long-sequence modelling. One key architectural element was the use of multiplicative gating units, in which the outputs of some neurons modulate the outputs of others. These multiplicative units are conceptually distinct from the additive attention mechanism later introduced for sequence-to-sequence models. Neural networks using multiplicative units were later called sigma-pi networks or higher-order networks. LSTM became the standard architecture for long sequence modelling until the 2017 publication of transformers. However, LSTM still used sequential processing, like most other RNNs. Specifically, RNNs operate one token at a time from first to last; they cannot operate in parallel over all tokens in a sequence. Modern transformers overcome this problem, but unlike RNNs, they require computation time that is quadratic in the size of the context window. The linearly scaling fast weight controller (1992) learns to compute a weight matrix for further processing depending on the input. One of its two networks has "fast weights" or "dynamic links" (1981). A slow neural network learns by gradient descent to generate keys and values for computing the weight changes of the fast neural network which computes answers to queries. This was later shown to be equivalent to the unnormalized linear transformer.